

# SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

## ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

|                  |                                                                                                                            |               |                                                |
|------------------|----------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| Course Code:     | DSA03                                                                                                                      | Course Title: | Natural Language Processing                    |
| CO Assessed:     | CO2 – Manipulation                                                                                                         | Assessment:   | Assessment Tool 1 – ANALYTICAL PROBLEM SOLVING |
| Weightage:       | 50% of CIA                                                                                                                 |               |                                                |
| CO5 Statement:   | Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3) |               |                                                |
| Student Name:    | P. Chola Lakshmi Narasimha                                                                                                 | Reg. No.:     | 192325021                                      |
| Section / Batch: |                                                                                                                            | Date:         |                                                |

### Code of Conduct

I P. Chola Lakshmi Narasimha/192325021 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: P. Chola Lakshmi Narasimha

### Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

| Section / Topic                                                                                     | Focus                                                                                                                                                                                  | Q Nos |
|-----------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| English Word Classes, Part of Speech Tagging                                                        | Morphological decomposition of connected, connecting, and connection; identification of roots, suffixes, inflectional vs. derivational forms, and normalization for document indexing. | Q1    |
| English Word Classes, Tagsets for English, Rule-Based Part of Speech Tagging                        | Morphological parsing of unhappy, happiness, and happily; identification of prefixes, suffixes, base forms, and semantic effects of derivational morphology.                           | Q2    |
| Counting Words, Part of Speech Tagging, English Word Classes                                        | Stemming and root extraction for played, player, and playing; handling grammatical variations and word formation changes for search preprocessing.                                     | Q3    |
| Finite-State Morphological Analysis, RuleBased Part of Speech Tagging, Transformation-Based Tagging | Finite-state parsing of writes, writing, and written; modeling regular and irregular verb inflections and root normalization.                                                          | Q4    |
| English Word Classes, Rule-Based Part of Speech Tagging, Counting Words                             | Porter Stemmer-based processing of relational, relation, and relate; application of derivational suffix removal and stem extraction for retrieval.                                     | Q5    |

## 1. Morphological Analysis Pipeline for Document Indexing

### Input Words

- connected
- connecting
- connection

### Step 1: Morphological Decomposition

| Word       | Root    | Suffix | Type         |
|------------|---------|--------|--------------|
| Connected  | Connect | -ed    | Inflectional |
| Connecting | Connect | -ing   | Inflectional |
| Connection | Connect | -ion   | Derivational |

### Step 2: Inflectional vs Derivational

- **Inflectional Morphology**
  - Does **not** change the word class.
  - Only changes tense, number, aspect, etc.
  - Example:
    - connect → connected
    - connect → connecting
- **Derivational Morphology**
  - Changes the meaning or grammatical category.
  - Example:
    - connect (verb)
    - connection (noun)

### Step 3: Normalization

All words are normalized to the common base form:

connected → connect

connecting → connect

connection → connect

### Morphological Parsing Output

| Input word | Root    | Affix | Classification | Normalized Output |
|------------|---------|-------|----------------|-------------------|
| Connected  | Connect | -ed   | Inflectional   | Connect           |
| Connecting | Connect | -ing  | Inflectional   | Connect           |
| Connection | Connect | -ion  | Derivational   | Connect           |

## Morphological Analysis Pipeline

Input Words



Tokenization



Identify Prefix/Suffix



Extract Root



Classify  
(Inflectional / Derivational)



Normalize to Base Form



Store in Document Index

## 2. Morphological Parsing for Sentiment Analysis

### Input Words

- unhappy
- happiness
- happily

### Rule-Based Decomposition

#### Word 1: unhappy

un + happy

- Prefix = un-
- Root = happy
- Suffix = None

Meaning:

Negation of "happy"

#### Word 2: happiness

happy + ness

Root = happy

Suffix = -ness

Forms a noun.

#### Word 3: happily

happy + ly

Root = happy

Suffix = -ly

Forms an adverb.

### Morphological Breakdown

| Word      | Prefix | Root  | Suffix | Classification |
|-----------|--------|-------|--------|----------------|
| Unhappy   | Un-    | Happy | -      | Derivational   |
| Happiness | -      | Happy | -ness  | Derivational   |
| Happily   | -      | Happy | -ly    | Derivational   |

### Semantic Consistency

All words express the same emotional concept related to **happy**.

unhappy → happy

happiness → happy

happily → happy

## Final Output

| Input     | Morphological breakdown | Normalized Root |
|-----------|-------------------------|-----------------|
| Unhappy   | Un+happy                | Happy           |
| Happiness | Happy+ness              | Happy           |
| Happily   | Happy+ly                | Happy           |

### 3. Stemming-Based Preprocessing Module

#### Input Words

- played
- player
- playing

#### Affix Stripping

**played**

played

remove -ed

Stem = play

**player**

player

remove -er

Stem = play

**playing**

playing

remove -ing

Stem = play

#### Classification

| Word           | Stem        | Affix       | Classification      |
|----------------|-------------|-------------|---------------------|
| <b>Played</b>  | <b>Play</b> | <b>-ed</b>  | <b>Inflectional</b> |
| <b>Player</b>  | <b>Play</b> | <b>-er</b>  | <b>Derivational</b> |
| <b>Playing</b> | <b>Play</b> | <b>-ing</b> | <b>Inflectional</b> |

#### Consistent Root Extraction

played

player

playing

↓

play

### Final Output

| Input Word | Stem | Classification |
|------------|------|----------------|
| played     | play | Inflectional   |
| player     | play | Derivational   |
| playing    | play | Inflectional   |

### Stemming Module

Input



Detect Suffix



Remove Affix



Extract Stem



Store "play"

## 4. Finite-State Morphological Parsing

### Input Words

- writes
- writing
- written

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### Finite-State Representation

#### writes

write  
|  
+ s  
|  
writes

---

#### writing

write  
|  
+ ing  
|  
writing

---

#### written

Irregular verb  
write  
|  
Irregular transformation  
|  
written

---

### State Transition Diagram

+-- s -----> writes  
/  
START → write  
\  
+-- ing -----> writing  
\  
+-- irregular ---> written

---

### Morphological Breakdown

| Word    | Root  | Affix           | Pattern   |
|---------|-------|-----------------|-----------|
| writes  | write | -s              | Regular   |
| writing | write | -ing            | Regular   |
| written | write | irregular (-en) | Irregular |



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## Regular vs Irregular

### Regular

- writes
- writing

### Irregular

- written
- 

## Normalized Output

All words become

Write

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## Final Table

| Input   | State Transition  | Morphological Breakdown | Normalized |
|---------|-------------------|-------------------------|------------|
| writes  | write → +s        | write + s               | write      |
| writing | write → +ing      | write + ing             | write      |
| written | write → irregular | write + en              | write      |

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## 5. Porter Stemmer-Based Preprocessing

### Input Words

- relational
  - relation
  - relate
- 

### Word 1: relational

#### Step 1

relational

Porter Rule:

-ational → ate

relational

↓

relate

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#### Step 2

relate

Remove trailing e

relat

Final stem:

relat

---

### Word 2: relation

#### Step 1

relation

Remove  
-ion  
Intermediate  
relat  
Final stem  
relat

---

### Word 3: relate

#### Step 1

relate  
Remove final e  
relat  
Final stem  
relat

---

### Step-by-Step Stemming

| Word       | Rule Applied   | Intermediate Form | Final Stem |
|------------|----------------|-------------------|------------|
| relational | -ational → ate | relate            | relat      |
| relation   | Remove -ion    | relat             | relat      |
| relate     | Remove final e | relat             | relat      |

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### Uniform Root Extraction

relational  
relation  
relate

↓

relat

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### Final Output

| Input Word | Stemming Steps              | Final Stem |
|------------|-----------------------------|------------|
| relational | relational → relate → relat | relat      |
| relation   | relation → relat            | relat      |
| relate     | relate → relat              | relat      |

| Faculty In-charge | Course Coordinator | Program Director |
|-------------------|--------------------|------------------|
|                   |                    |                  |
| Signature: _____  | Signature: _____   | Signature: _____ |
| Date: _____       | Date: _____        | Date: _____      |

## Rubrics for Calculation

| Numerical Problems             |                              |                                                                                   |                                           |                                            |                                               |                   |                |
|--------------------------------|------------------------------|-----------------------------------------------------------------------------------|-------------------------------------------|--------------------------------------------|-----------------------------------------------|-------------------|----------------|
| Criteria                       | Weightage                    | Level 4<br>(Exemplary)                                                            | Level 3<br>(Proficient)                   | Level 2<br>(Developing)                    | Level 1<br>(Beginning)                        |                   |                |
| Problem Understanding          | 20%                          | Clearly interprets problem, identifies all parameters and requirements accurately | Understands problem with minor gaps       | Partial understanding; misses key elements | Misinterprets or unable to understand problem |                   |                |
| Method Selection               | 20%                          | Selects most appropriate method/formula with strong justification                 | Selects correct method with minor errors  | Inappropriate or partially correct method  | Unable to select method                       |                   |                |
| Solution Process               | 25%                          | Logical, systematic steps with correct approach throughout                        | Mostly correct steps with minor mistakes  | Disorganized steps; multiple errors        | No clear process                              |                   |                |
| Accuracy of Calculation        | 20%                          | All calculations accurate; correct final answer                                   | Minor calculation errors                  | Frequent errors; incorrect final answer    | No valid calculations                         |                   |                |
| Interpretation of Results      | 15%                          | Clearly interprets results with units and engineering relevance                   | Basic interpretation with minor omissions | Limited or unclear interpretation          | No interpretation                             |                   |                |
| <b>Q. No. / Criteria Level</b> | <b>Problem Understanding</b> | <b>Method Selection</b>                                                           | <b>Solution Process</b>                   | <b>Accuracy of Calculation</b>             | <b>Interpretation of Results</b>              | <b>Sub. Total</b> | <b>Total %</b> |

|   |  |  |  |  |  |  |  |
|---|--|--|--|--|--|--|--|
| 1 |  |  |  |  |  |  |  |
| 2 |  |  |  |  |  |  |  |
| 3 |  |  |  |  |  |  |  |
| 4 |  |  |  |  |  |  |  |
| 5 |  |  |  |  |  |  |  |